

The effect of cigarette smoke, nicotine, and carbon monoxide on the permeability of the arterial wall

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The association between cigarette smoking and the development of atherosclerosis is well established, but the mechanism that makes cigarettes such a potent "risk factor" is not understood. There is normally a constant insudation of plasma macromolecules into the arterial wall. Fibrinogen and lipids are two of the large molecules involved in atherosclerosis. Therefore we studied the effect of cigarette smoke, nicotine, and carbon monoxide on the permeability of the canine arterial wall to ^{125}I -labeled fibrinogen. The results show that inhaled cigarette smoke significantly and rapidly increases the permeability of the arterial wall to fibrinogen and that this effect can be produced with carbon monoxide alone but not with intravenous nicotine. (*J VASC SURG* 1988;7:139-52.)

The arterial wall and its endothelial lining were once thought to be an impermeable tube, but it is now evident that there is a continuous ultrafiltration of plasma constituents through the arterial wall. Ross and Glomset^{1,2} have put forward the "response-to-injury" hypothesis, which links endothelial damage and smooth muscle cell proliferation to the incorporation of lipids and fibrin into the arterial wall. If this hypothesis is correct, the most likely part of the sequence on which cigarette smoke could exert its atherogenic effect is through endothelial damage.

Woolf and Wilson Holt³ have shown that after a single brief exposure to cigarette smoke, endothelial cells from rabbit arteries become swollen and that longer exposure causes the appearance of sharply demarcated holes, which may be the stomata of active pinocytotic vessels; however, the endothelium maintains its integrity. These latter changes were particularly obvious in rats that had been subjected to smoke from two cigarettes a day for 10 days.³

If cigarette smoke alters the appearance of endothelial cells, it probably affects the permeability of the endothelium and allows the influx of lipids and other plasma macromolecules into the arterial wall. The experiments described in this article were designed to test this hypothesis.

MATERIAL AND METHODS

All animal experiments and care complied with Home Office regulations.

A polystyrene capsule has been developed that can be placed around the femoral artery of a dog.⁴ Once in place, the capsule provides a means of collecting molecules at the adventitial surfaces after they have crossed the arterial wall (Fig. 1).

The capsule is injection-molded and in two parts. At each end of each half is a semicircular channel. After polystyrene cement is applied to the edge of each half, and the locating pins in the bottom half are aligned with corresponding holes in the top half, the two halves are held together around the artery with modified artery forceps until the cement sets. When the two halves of the capsule are glued together, the semicircular channels at either end form two tunnels, 4 mm in diameter at the proximal end and 3 mm at the distal end. The tunnels are 3 mm long and thus grip the artery sufficiently to form a watertight seal without causing undue compression of the vessel. If necessary, larger vessels (6 mm diameter or larger) can be accommodated by reaming out the channels before the capsule is positioned.

In the top half of the capsule are two ports through which silicone tubing (external diameter 2 mm) is pulled. The tubing in the proximal port is connected to a calibrated 2 ml syringe filled with warmed (37°C) Ringer's solution. The tubing in the distal port is left open to the atmosphere and thus acts as an air vent. The capsule can then be filled with Ringer's solution, which bathes the entire outer surface of the artery that lies within the capsule.

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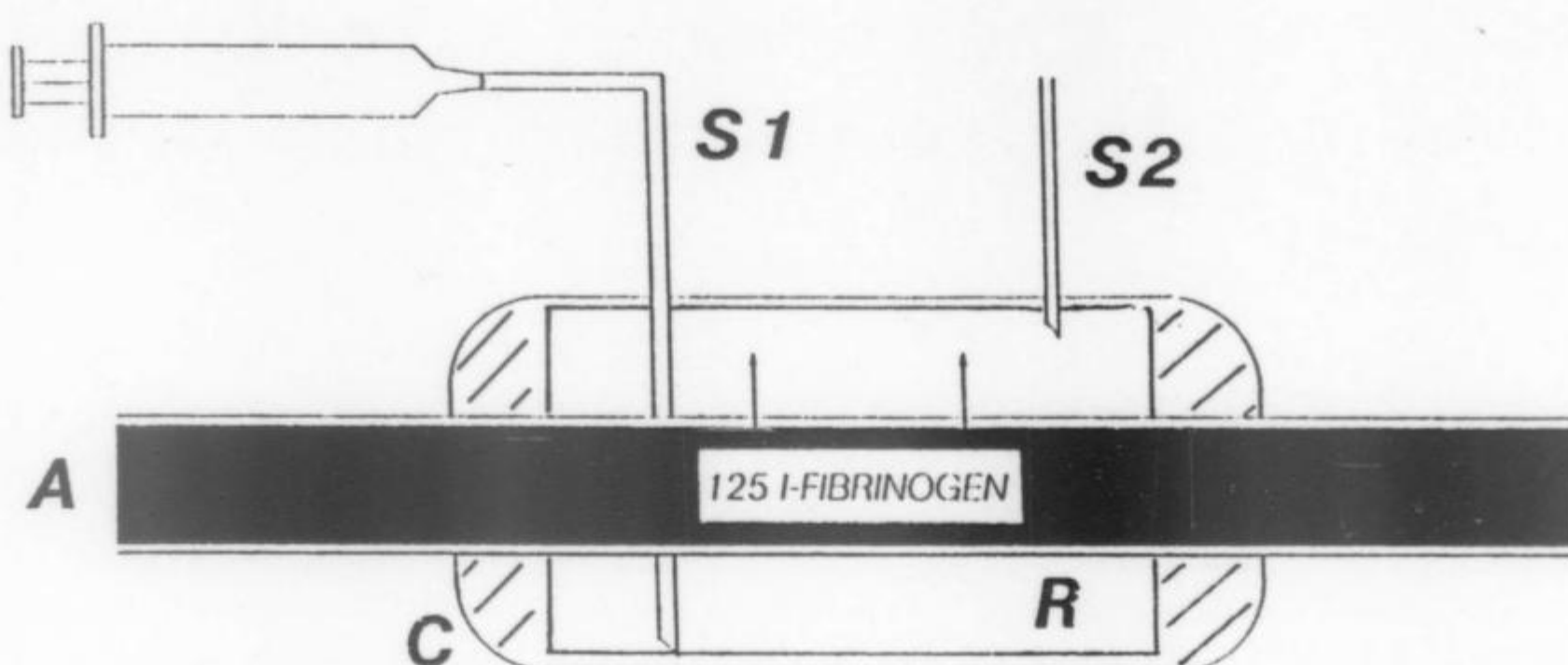


Fig. 1. The arterial capsule (see text). When the two halves of the polythene capsule (C) are glued together around the artery (A), they form an enclosed space that can be filled with Ringer's solution (R). Silicone rubber tubes (S1 and S2) allow the capsule to be emptied and refilled at regular intervals. The quantity of radioactive fibrinogen that passes from the blood stream through the arterial wall into the capsule fluid can be measured and the permeability coefficient of the artery calculated.

The inner ends of the silicone rubber tubes are cut to different lengths; the distal tube is cut flush with the inner surface of the capsule to allow all the air to be expelled. The proximal tube is cut obliquely and long enough to reach the bottom of the capsule beneath the artery.

To measure permeability, one vial of ^{125}I -labeled fibrinogen (1.2 mg, 4.1 MBq) (Amersham Laboratories, Amersham, Bucks, England), was injected intravenously via a peripheral vein after each capsule was filled with Ringer's solution. Measurement of the quantity of radioactive fibrinogen that had crossed the arterial wall was made by withdrawing the Ringer's solution at 30-minute intervals with a 2 ml syringe attached to the proximal Silastic tube, transferring it into a plastic test tube and placing it in a well scintillation counter (Panax Ltd., Redhill, Surrey, England) where the counts for 300 seconds were recorded with a counter/ratemeter (MS 310, J & P Engineering, Reading, England). Two recordings were made with each sample. The capsule was refilled immediately with a fresh volume of Ringer's solution with a fresh syringe. The exact volume used to refill the capsule and the volume subsequently emptied 30 minutes later were recorded.

The plasma radioactivity of arterial blood was also measured every 30 minutes with 10 ml of blood withdrawn from the arterial line and collected in a lithium-heparin bottle. After centrifugation for 10 minutes, 2 ml of supernatant plasma was placed in a plastic test tube and the radioactivity counted for 100 seconds in the well counter. Two recordings were

made of each sample. Background readings from the well counter were made before and after each experiment.

Calculation of the permeability coefficient of the arterial wall. The amount of labeled fibrinogen crossing the arterial wall will in part depend on the concentration of labeled fibrinogen in the plasma. This varied from one dog to another with differences in body weight and plasma volume and the variable activity of the isotope dependent on the time interval between the reference date and the time the fibrinogen was used. Therefore, it was not feasible to compare the actual radioactivity in the Ringer's solution from one dog to another. Hence a measurement of permeability, which we have termed the permeability coefficient (PC), had to be calculated to compensate for these variable factors.

After the background count was subtracted, the counts in the Ringer's solution were converted to counts per milliliter per minute per square centimeter of artery contained within the capsule by means of the following equation:

$$C \text{ Ringer's (ml}^{-1} \cdot \text{min}^{-1} \cdot \text{cm}^{-2}) = C \times \frac{60}{300} \times \frac{1}{V} \times \frac{1}{SA}$$

where C is actual counts of Ringer's solution in 300 seconds minus background, V is volume of Ringer's solution counted, and SA is surface area of the artery ($2\pi \times \text{radius [cm]} \times \text{length [cm]}$).

To relate the radioactivity of the Ringer's solution to that of the plasma, both samples should be taken and measured at the same time. Logistic problems

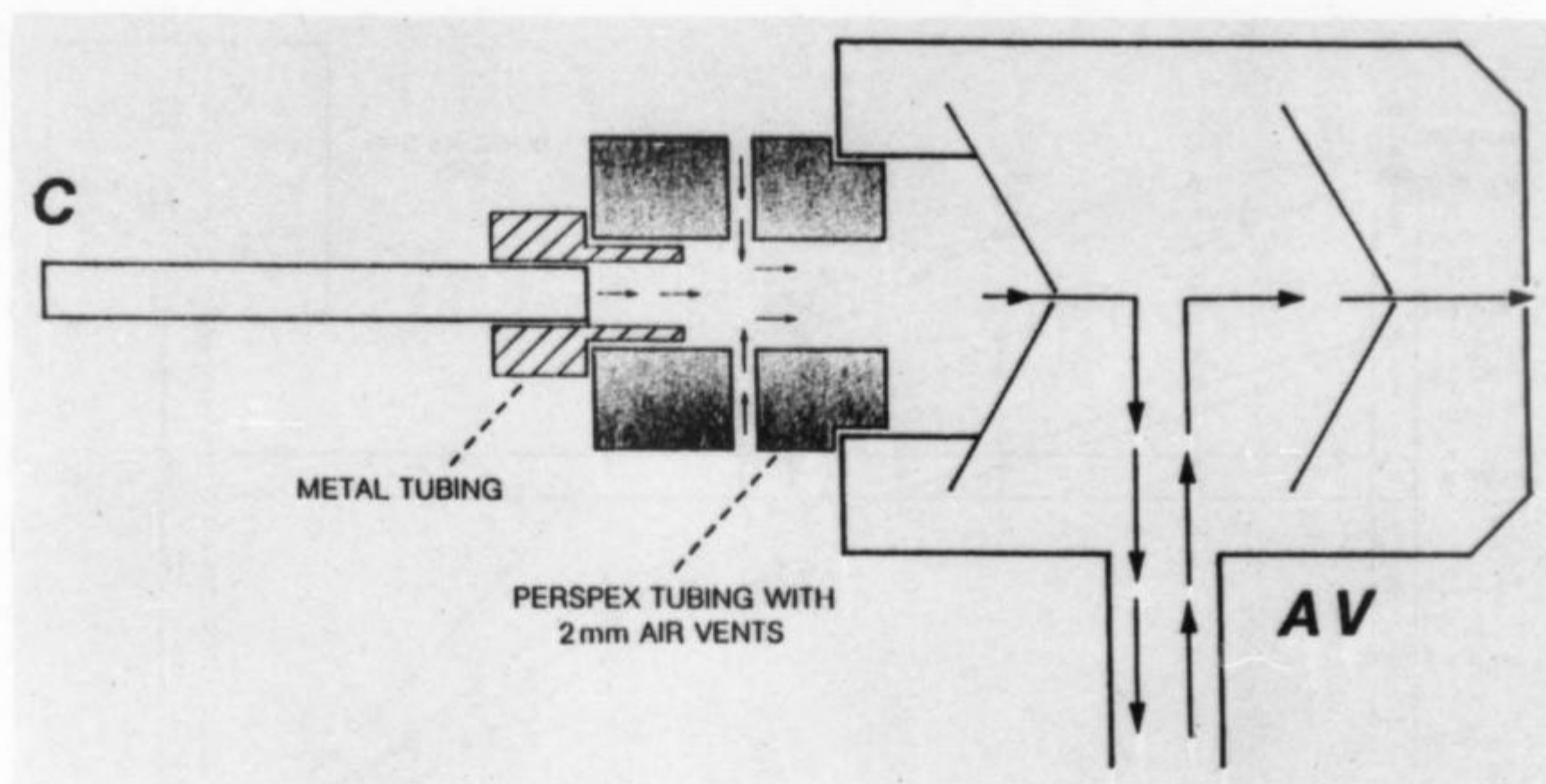


Fig. 2. The attachment to the Ambu valve (AV) that allows the inhalation of cigarette smoke from the lighted cigarette (C) and room air.

prevented this being done so the plasma sample was taken 15 minutes after the intravenous injection of the fibrinogen and thereafter every 30 minutes, thereby enabling the measurement of the plasma and Ringer's solution radioactivity to be made with a single well counter. The plasma radioactivity in counts per milliliter per minute was then plotted against time to allow a linear regression analysis to be performed to determine the "biologic" decay of the labeled fibrinogen. The radioactivity of the plasma at the time that the Ringer's solution was collected and measured was then read off from the regression line.

The PC of the artery wall was then calculated with the equation:

$$PC \text{ (units} \cdot \text{cm}^{-2}) = \frac{C \text{ Ringer's (ml}^{-1} \cdot \text{min}^{-1} \cdot \text{cm}^{-2})}{C \text{ plasma (ml}^{-1} \cdot \text{min}^{-1})}$$

The background counts in the well counter at the start and at the end of each experiment ranged between 1.5 and 11 cpm. The mean reduction of plasma radioactivity during a 3-hour experiment was 20% (range 14.2% to 27.2%).

Preparation and physiologic monitoring. All dogs were anesthetized with intravenous sodium pentobarbitone (30 mg/kg) (Sagatal, M & B Veterinary Products, Dagerham, Essex, England) and maintained with additional 120 mg intravenous boluses as required. Most dogs only needed between one and three 120 mg boluses during the experiments, which lasted 5 to 6 hours. After induction of anesthesia, a fine (19-gauge) catheter was inserted into the cephalic vein and 0.9% saline solution was

given at a rate of 200 ml/hr. Central venous pressure was monitored through a catheter inserted via the jugular vein and arterial pressure was monitored via a catheter in either a carotid artery or a brachial artery, both connected to pressure transducers (model 840, Sensoror A/S, Horten, Norway) and an amplifier/display unit (S & W Medico Teknik A/S, Albertslund, Denmark). The central venous and arterial pressures were recorded every 30 minutes throughout the experiment. A cuffed endotracheal tube (7.5F, Portex Inc., Wilmington, Mass.) was inserted and all dogs were allowed to breathe spontaneously.

Incisions were made in both limbs, from the inguinal ligament to the midportion of the thigh. The superficial femoral arteries were mobilized and all branches were ligated with 6-0 silk sutures. Care was taken to keep the vessels moist throughout the dissection with warmed saline solution (0.9%). After the diameter of the artery was measured with calipers, a capsule was placed around each artery.

All dogs were killed at the end of each experiment.

Control animals. All dogs in this group were intubated and breathed room air. A capsule was placed around each femoral artery as described and permeability measurements were made over a period of 3 hours.

Smoke inhalation. Inhalation of cigarette smoke was achieved by means of the apparatus shown in Fig. 2. The cuffed portex endotracheal tube was connected to an Ambu valve (Ambu International, Glostrup, Denmark). A polystyrene and metal cigarette holder was inserted into the inlet side of the valve.

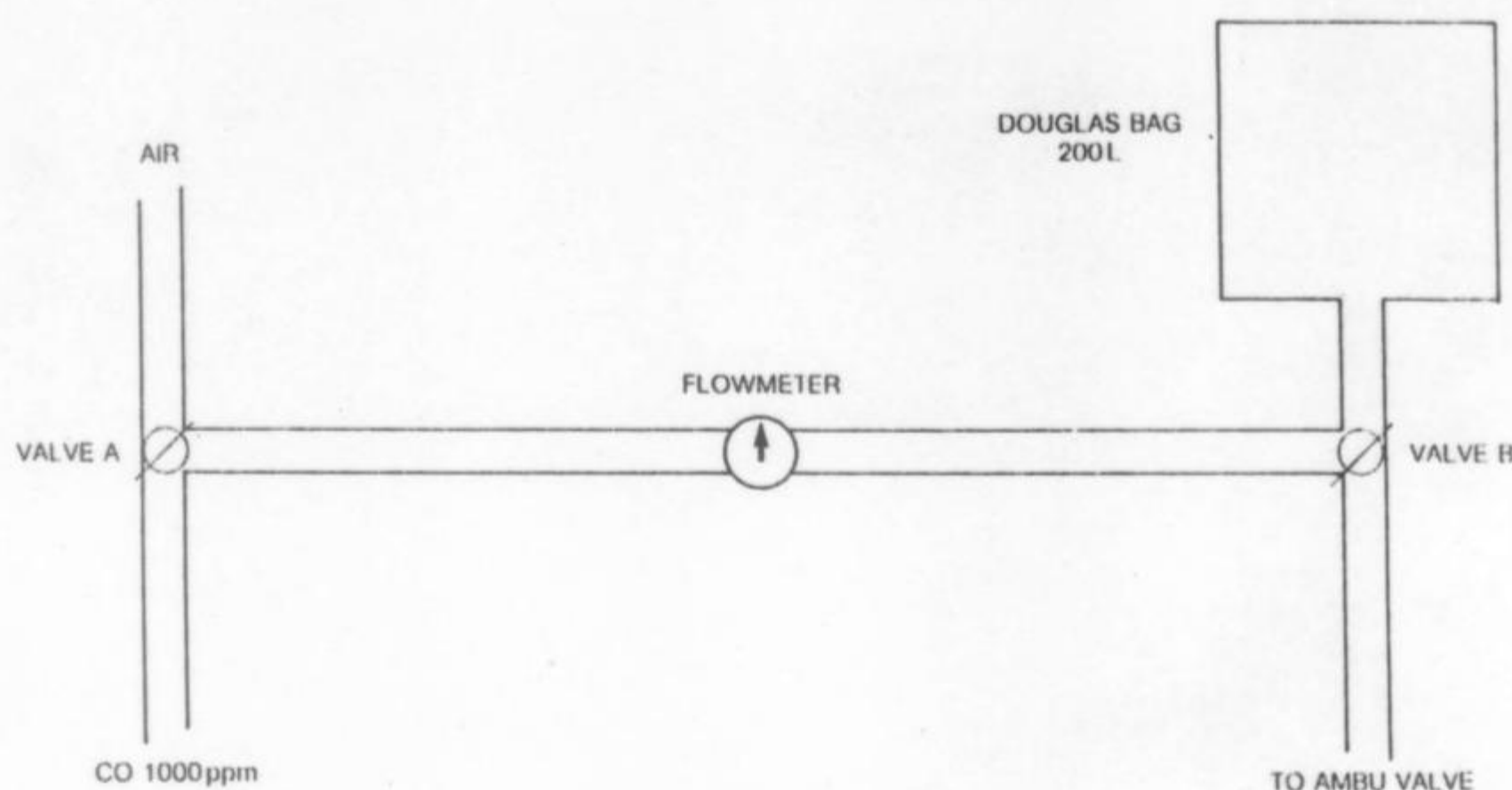


Fig. 3. The circuit diagram of the apparatus used for mixing variable quantities of carbon monoxide and room air.

After the dogs inhaled the smoke of four cigarettes an hour for 2 hours, radioactive fibrinogen was injected intravenously and permeability measurements were started.

Inhalation of smoke continued throughout the 3-hour permeability experiment. At the end of a 5-hour period, the dog had "smoked" 20 cigarettes. The brand of cigarettes used were graded as "high tar and high nicotine content."

Arterial blood samples were taken into 5 ml syringes containing heparin before smoke inhalation started and every 30 minutes throughout the experiment. The samples were stored in ice immediately and later analyzed for percentages of carboxyhemoglobin (COHb) and oxyhemoglobin (O₂Hb) with a CO oximeter analyzer modified for use with dog blood (IL 282, Instrumental Laboratory, Lexington, Mass.).

Nicotine infusion. Nicotine base (25 mg) was diluted in 50 ml of 0.9% saline solution (9 gm/L) and infused at a rate of 10 ml/hr for 90 minutes before permeability measurements were started and then were continued throughout the 3-hour experiment. This provided an infusion of approximately 2.5 µg/kg/min.

The plasma nicotine levels (performed by Dr. C. Feyerabend of the Guy's Hospital Poisons Unit) were measured with a flame-ionizing gas-chromatography technique with nitrogen-sensitive detectors. This method is sensitive to 0.1 ng/ml.⁵ Nicotine levels were measured 90 and 270 minutes after the infusion was started.

Carbon monoxide (CO) inhalation. The apparatus used to administer CO is shown in Fig. 3. CO (1000 ppm, Morgans Ltd., Rainham Kent, England) was diluted with air to produce a concentration of 400 ppm of CO. The dogs inhaled CO for 2 hours before the permeability measurements were begun and continued to inhale CO until the end of the experiment 3 hours later. COHb levels were measured every 30 minutes with the CO oximeter analyzer.

Histologic examination. At the end of each experiment, that portion of the superficial femoral artery that lay within the capsule was removed and prepared for light microscopy (LM) and electron microscopy (EM). Specimens for LM were obtained by cutting transverse sections at three points along the artery and placing the specimens in 10% formaldehyde. Sections (4 µm) were cut and stained with hematoxylin-eosin and elastin-van Gieson technique.

EM sections were prepared by cutting transverse sections (1 mm thick) of the artery with a razor blade and then, from these sections, cutting 1 mm cubes from three points equidistant from each other around the artery. The three cubes were immediately placed in 2% glutaraldehyde for 24 hours and then postfixed in 1% osmium tetroxide, dehydrated, and embedded in araldite. Sections (80 nm) were cut on a microtome (LKB Instrument GMBH, Graefelfing, West Germany), stained with uranyl acetate and lead citrate, and examined in a transmission electron microscope (model 201, Philips Electronic Instruments

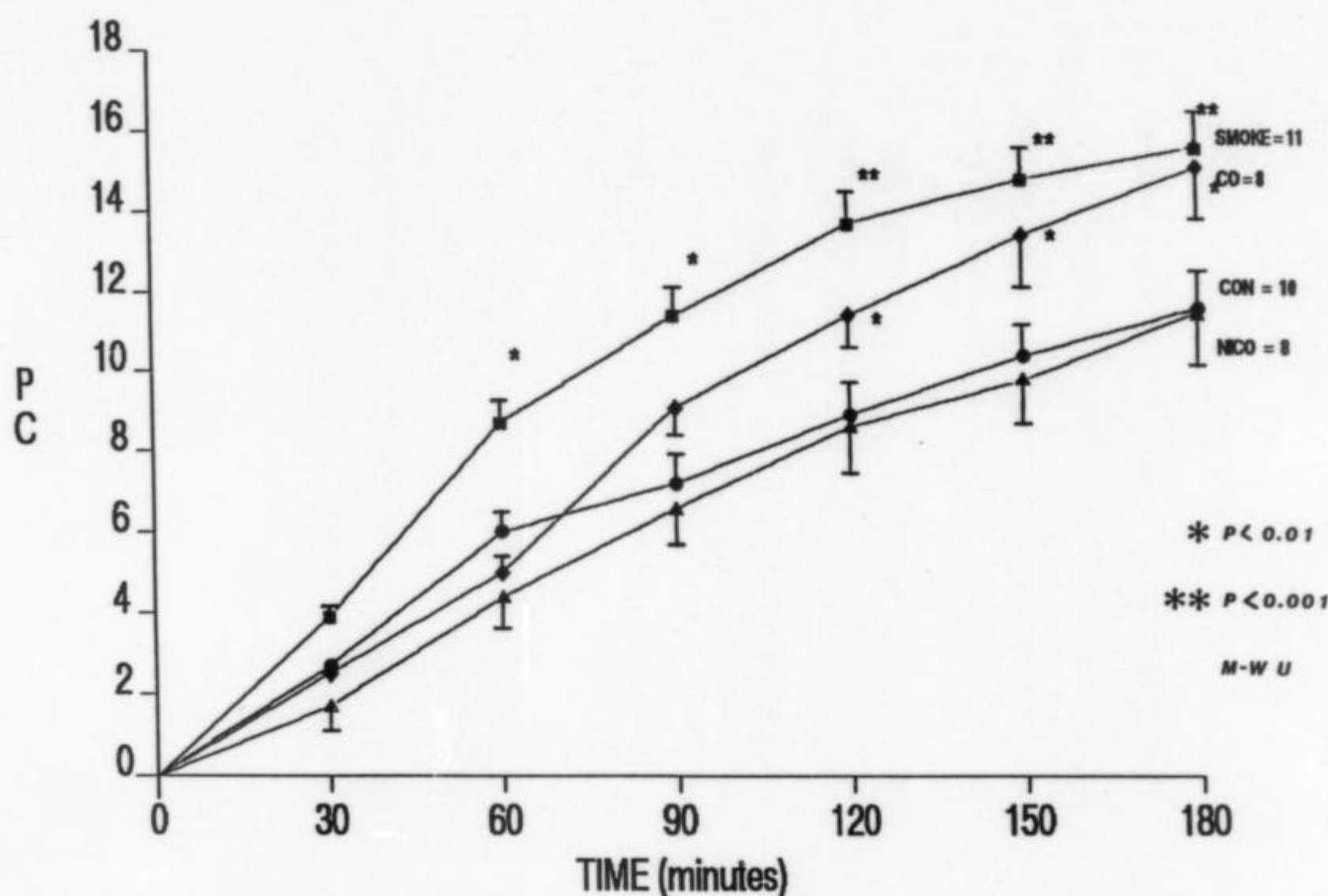


Fig. 4. The permeability coefficient (PC units, $\text{cm}^2 \times 10^{-4}$) of the control (CON), smoking (SMOKE), carbon monoxide (CO), and nicotine infusion (NICO) experiments. Means and standard deviations are presented. * = statistical significance between smoking and controls after 180 minutes ($p < 0.01$). ** = statistical significance between carbon monoxide inhalation and control after 180 minutes ($p < 0.001$).

Inc., Mahwah, N.J.) at 60 kV. All the histologic specimens were examined by one pathologist who reported each specimen without knowing the nature of the experiment.

RESULTS

Control studies. The permeability of 10 arteries in six control dogs was studied. The mean weight of the dogs in this group was 29.5 ± 2.8 kg. The mean total dose of Sagatal required from the induction of anesthesia to the end of the experiment was 0.13 ± 0.02 mg/kg/min. The mean duration of these experiments was 318 ± 22 minutes. The mean external diameter of the artery was 6.2 ± 0.4 mm. The resting blood COHb and O₂Hb levels were $1.6\% \pm 0.1\%$ and $86\% \pm 2.4\%$, respectively.

The PC (in units per square centimeter) of the 10 control arteries increased steadily throughout the experiment (Fig. 4). Three hours after the injection of fibrinogen, the PC was 11.6 ± 0.9 . The arterial blood pressure did not change significantly (Fig. 5).

Smoke inhalation studies. Eleven arteries of seven dogs were studied. The mean weight of the dogs was 29.3 ± 4.3 kg. The mean diameter of the artery was 5.2 ± 0.7 mm. This group of experiments lasted slightly longer than the control group

with a mean duration of 362 ± 36 minutes but the total dose of sodium pentobarbitone required was 0.11 ± 0.01 mg/kg/min, not significantly different from that of the control group.

The mean COHb increased from a resting level of $1.5\% \pm 0.1\%$ to a mean level of $7.4\% \pm 0.9\%$ after the smoke of eight cigarettes was inhaled over a period of 2 hours, at which point permeability measurements were started. After another 3 hours of smoke inhalation the mean COHb level had increased to $12.8\% \pm 0.8\%$. These levels are within the range found in human smokers. There was a 7% decrease in O₂Hb during the experiment. The levels of plasma nicotine at the end of the experiment in two of the dogs subjected to smoke was 47.6 and 54 ng/ml, levels similar to those found in humans who smoke.⁶

The changes in arterial wall permeability are shown in Fig. 4. The PC increased steadily throughout the experiment at a rate greater than and to a level significantly higher than that in the control studies. At the end of the experiment the PC of the smoke inhalation and control groups was 15.6 ± 0.9 and 11.6 ± 0.9 , respectively ($p < 0.001$).

There was a moderate correlation between the PC and the %COHb of the animals subjected to

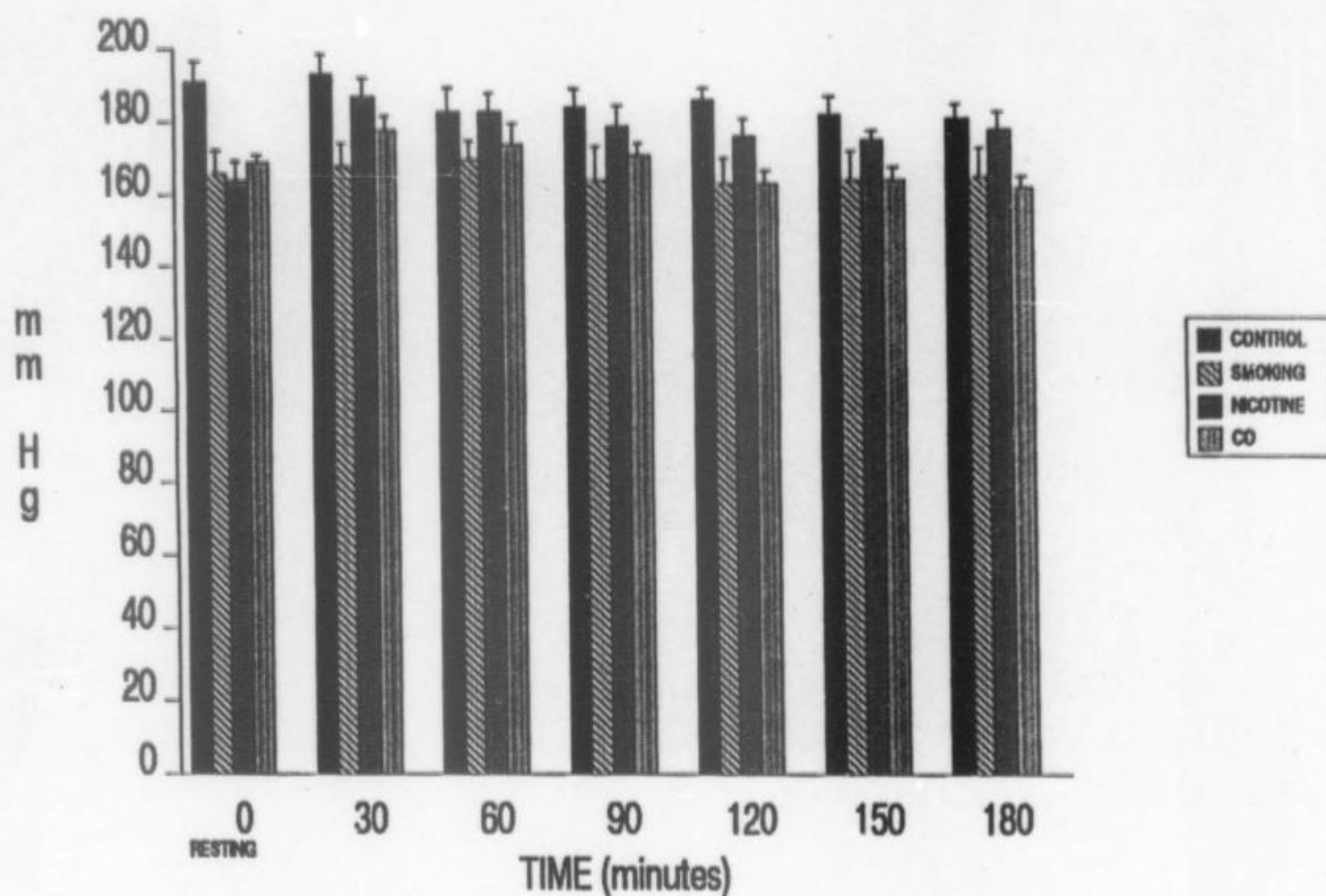


Fig. 5. Mean blood pressures of control, smoking, carbon monoxide inhalation, and nicotine infusion groups throughout the 180-minute permeability studies.

Table I. Plasma nicotine levels during nicotine infusion

Experiment No.	Plasma nicotine levels	
	at 90 min (ng/ml)	at 270 min (ng/ml)
N14	55.1	56.0
N15	77.9	62.9
N16	53.2	22.5
N17	50.7	53.7

smoke ($r = 0.58$, $p < 0.001$). The relationship was approximately a 1.5 to 2 unit increase in permeability for each 1% change of COHb.

Nicotine infusion. Eight arteries of four dogs were studied. The mean weight of the dogs in this group was 30.7 ± 1.8 kg. The mean diameter of the arteries was 5.5 ± 0.4 mm. The mean duration of the experiments was 308 ± 15 minutes and the mean total anesthetic requirement was 0.12 ± 0.02 mg/kg/min.

The plasma nicotine levels of the four dogs are shown in Table I. The mean plasma nicotine level at 90 minutes was 59.2 ± 12.5 ng/ml and at 270 minutes was 48.8 ± 18.0 ng/ml. These levels are within the range found in human smokers⁶ and are similar

to the plasma nicotine measurements found in the two dogs in the group subjected to smoke in which these measurements were made.

The PC increased steadily throughout the 3 hours after the fibrinogen injection at a rate identical to that observed in the control studies (Fig. 4). There was a slight increase in arterial blood pressure and an increase in heart rate, which had stabilized by the time the fibrinogen was injected (Fig. 6).

The blood nicotine level remained relatively constant throughout each experiment so it was not possible to see whether a relationship existed between the blood nicotine level and arterial wall permeability.

CO inhalation. Eight arteries of four dogs were studied. The mean weight of the dogs was 29.8 ± 2.9 kg. The mean diameter of the arteries was 5.4 ± 0.4 mm. The mean duration of each experiment was 326 ± 8 minutes and the mean dose of sodium pentobarbitone given during the experiment was 0.13 ± 0.01 mg/kg/min. Each dog inhaled CO at 400 ppm for 2 hours before the injection of fibrinogen and the start of permeability measurements and continued to inhale CO for another 3 hours. The mean volume of CO inhaled was 1000 L (900 to 1200 L). In the first 2 hours the COHb increased from a resting level of $2.0\% \pm$

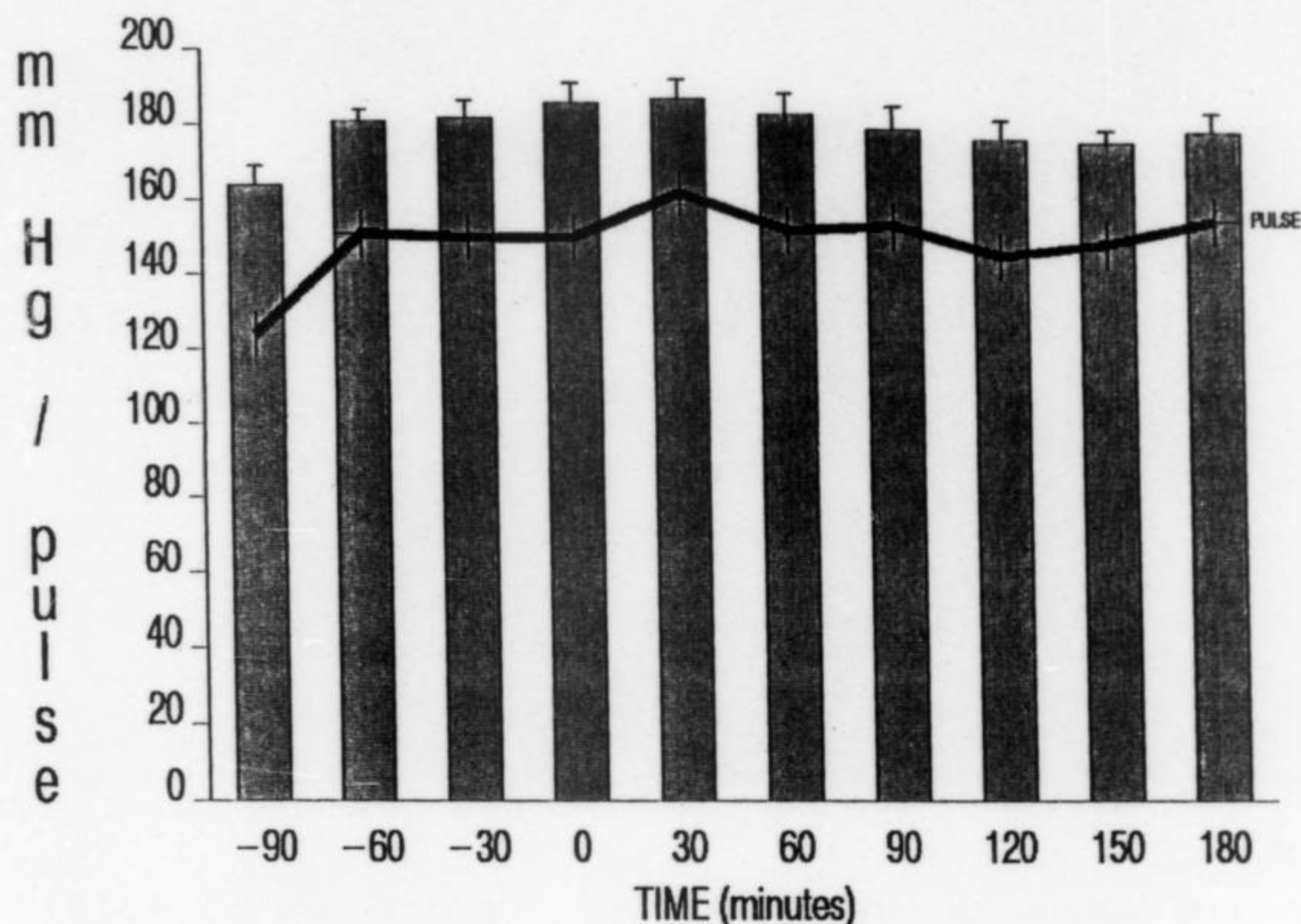


Fig. 6. Mean ($\pm$ SD) blood pressure and pulse rates during nicotine infusion. After an initial increase, both remained constant.

0.2% to $10.0\% \pm 0.8\%$. At the end of the experiment, after 5 hours of CO inhalation, the mean level had increased to $19.8\% \pm 1.4\%$. The % O₂Hb decreased from a mean of $90.0\% \pm 2.9\%$ at the start of the experiment to $76\% \pm 2.2\%$ at the end. This represents a 14% decrease of the O₂Hb level, a minor degree of hypoxia. The arterial blood pressure and heart rate did not alter significantly.

The PC steadily increased throughout the experiment at a rate and to a degree that was significantly greater than that of the control group ($p < 0.01$) (Fig. 4) but almost identical to that of the smoke inhalation group. At the end of the experiment the mean PC of this group was 15.1 ± 1.2 whereas that of the control groups was 11.6 ± 0.9 .

There was a good correlation between the blood COHb level and the PC ($r = 0.74$, $p < 0.001$), the PC increasing by approximately 2 units for every 1% increase in COHb. This correlation is similar to but statistically better than that observed in the smoke inhalation group.

Histologic examination. LM revealed no evidence of damage to the arterial wall in any of the groups. The endothelial layer and internal elastic lamina appeared to be intact and normal in every artery

studied. Transmission electron microscopy confirmed that the endothelial cells were normal and firmly attached to the internal elastic lamina and there was no evidence of widening of the interendothelial cell junctions (Figs. 7 and 8).

DISCUSSION

Atherosclerosis is essentially a human disease, in which lipids and fibrinogen accumulate in the sub-intimal layers of large elastic arteries and medium-sized muscular arteries. Although not a new disease, the prevalence and importance of atherosclerosis in the western world have increased dramatically since the turn of the century, a change that coincided with the invention of cigarette-making machines, which made the smoking of tobacco both cheap and convenient. Until this time, 80% of tobacco was taken as snuff or smoked in a pipe.

Smith and Slater⁷ have shown that the intima contains plasma lipoproteins in concentrations linearly related to their concentration in the plasma. It is probable that these molecules are removed from the outer layers of the artery by the vasa vasorum and lymphatics,⁸ although there is scant published information about the anatomy or function of the

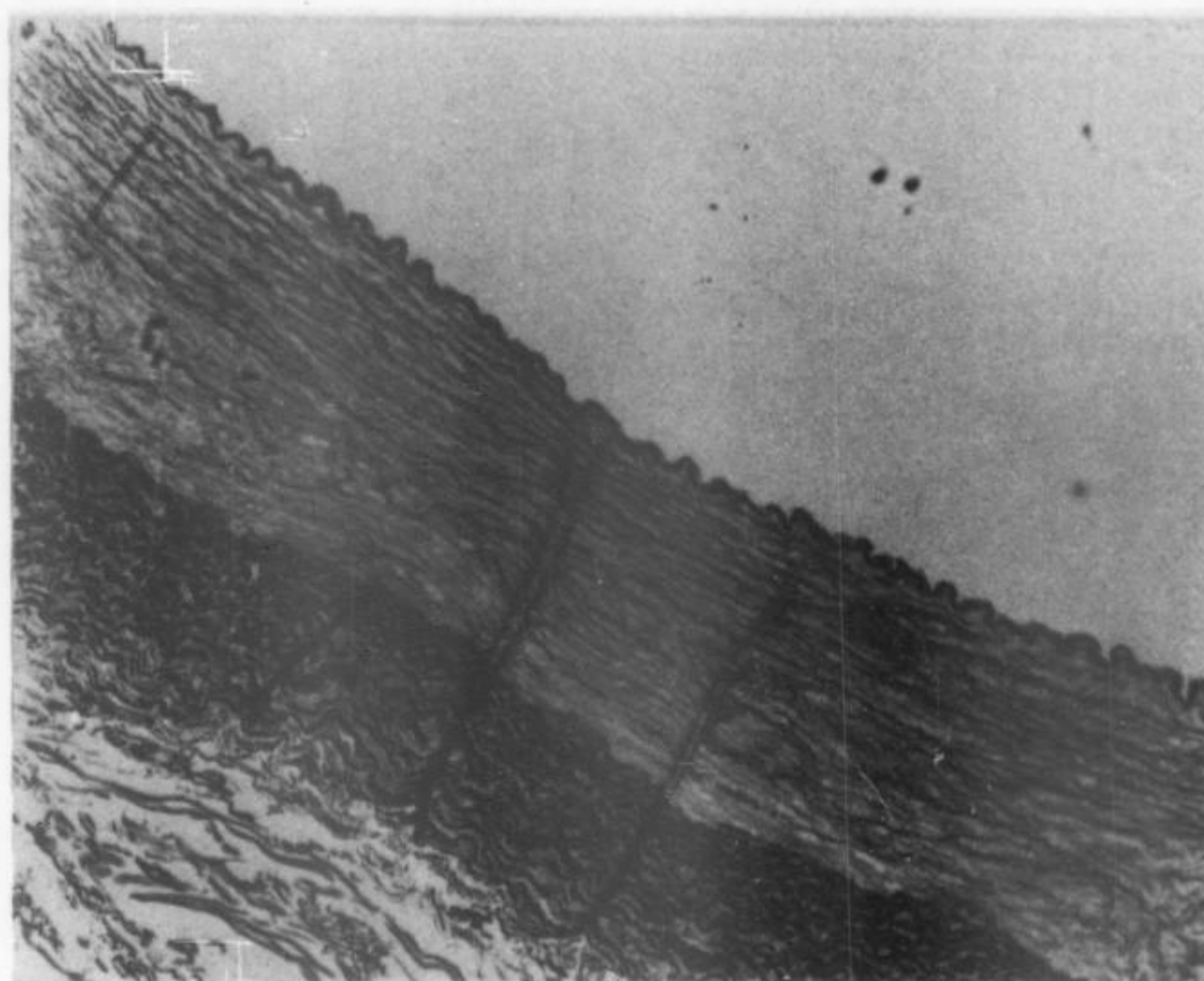


Fig. 7. Light transmission photomicrograph of the artery wall at the end of a 5-hour period of inhaling cigarette smoke. Endothelium and all other layers appear to be undamaged and there is no inflammatory response.

lymphatic and venous drainage of the arterial wall. The endothelial layer probably provides a selective barrier that regulates the passage of molecules because there is evidence that endothelial damage increases the permeability of the arterial wall; Bondjers and Bjorkerud⁹ found that tritiated cholesterol was incorporated into the wall of an artery with damaged endothelium, at a concentration 15 times that found in undamaged arteries. Although it is unlikely that such severe endothelial damage is the initiating event of atherosclerosis, it is not unreasonable to assume that the arterial wall suffers many minor injuries, ranging from an alteration in the constituents and health of the intimal cells to frank desquamation. Although the less severe forms of injury may have no morphologic manifestation, they may be of physiologic significance.

The dangers of smoking tobacco have been appreciated for many years. However, it was not until the 1950s that sound epidemiologic evidence accrued to demonstrate the strong association between smoking and the development of atheroma. The Framingham study is the best known prospective investigation that established the risk factors for coronary heart disease and peripheral vascular disease. Subjects (N = 5209) from Framingham, Massachusetts were examined biannually from 1949 for a period of 16 years.¹⁰ Five factors were shown to be predictors of

atherosclerosis: hypertension, a serum cholesterol level above 297 mg/100 ml (7.7 mol/L), glucose intolerance, left ventricular hypertrophy on electrocardiography, and smoking. Of these five risk factors, smoking was the most important in the development of intermittent claudication.¹⁰ To the vascular surgeon the association between smoking and atherosclerosis is obvious. In 1962 Eastcott¹¹ noted the rarity of lower limb ischemia in nonsmokers and more recently Laing et al.¹² found that more than 97% of their patients with peripheral vascular disease were smokers.

Although the association between cigarette smoking and the development of atherosclerosis is well established, the mechanism that makes smoking such a potent risk factor is not known. There is good evidence to suggest that cigarette smoke damages the endothelium. Asmussen and Kjeldsen¹³ examined the endothelium of umbilical arteries from neonates born to smoking and nonsmoking mothers with both SEM and transmission electron microscopy techniques and detected evidence of early endothelial damage, with endothelial cell swelling and extensive subendothelial edema, in the smoking group. With a similar human model, Bylock et al.¹⁴ found ultrastructural evidence of endothelial cell injury in the uterine arteries of smoking women when compared with those of nonsmokers.

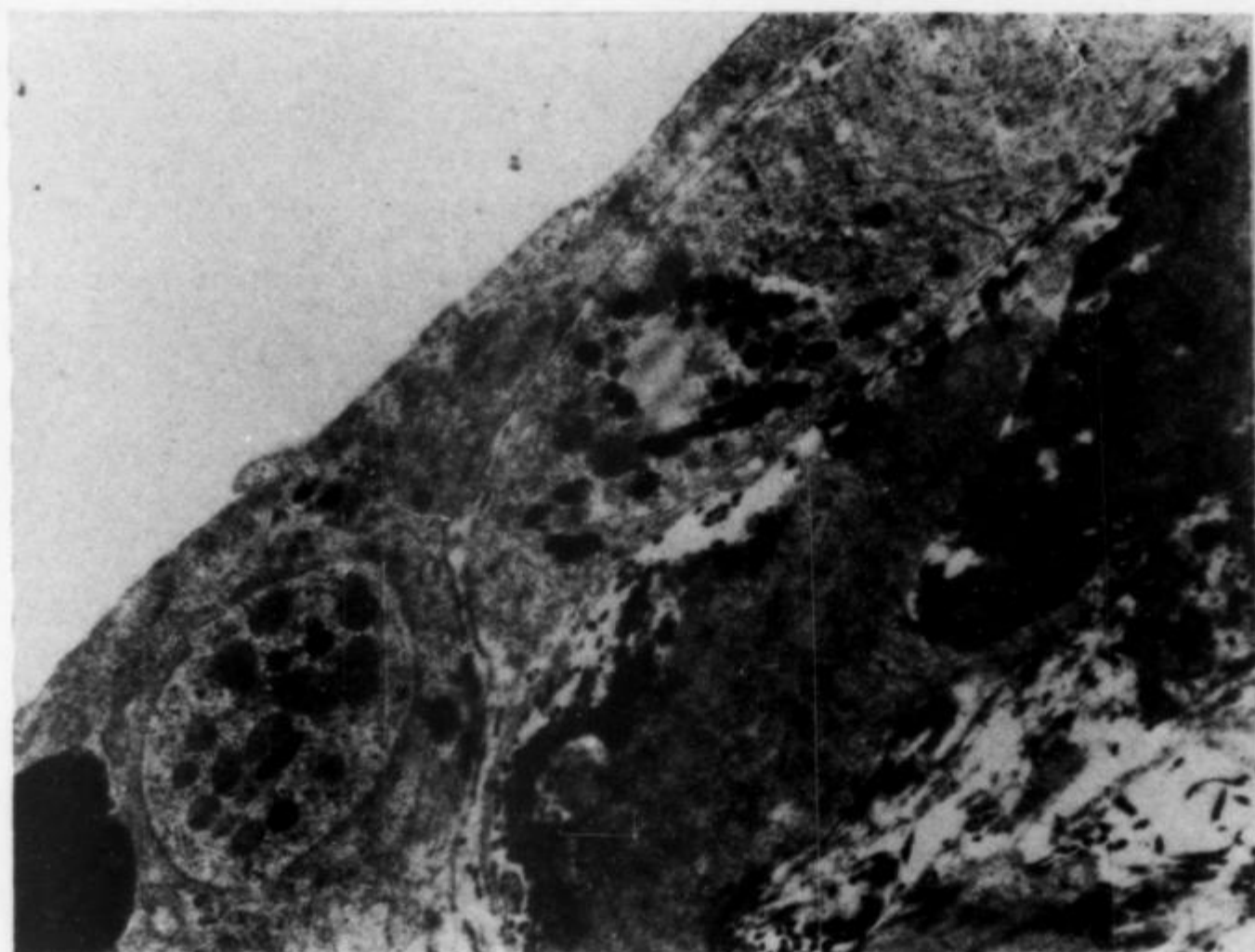


Fig. 8. Electron photomicrograph of the intimal cells at the end of a 5-hour period of inhaling cigarette smoke. The cell wall, its structures, and the interendothelial cell junctions are entirely normal.

Nicotine is the main alkaloid in cigarette smoke. The nicotine content of individual cigarettes varies between 0.5 to 3.0 mg. Plasma nicotine levels in cigarette smokers may increase to a level between 4 and 70 ng/ml,¹⁵ the level varying according to the person's smoking pattern. Intravenous nicotine has been shown to increase serum free fatty acid concentration in dogs but not to alter serum cholesterol or triglyceride levels.¹⁶ A desquamating effect of nicotine on vascular endothelium has been shown in rats when it was given intravenously or orally in doses between 12.5 and 25.0 mg/kg.¹⁷ The oral administration of nicotine (2.4 mg/kg/day) for 25 weeks has been shown to cause extensive morphologic changes in the endothelial cells of the rabbit aortic arch when assessed with the technique of Evans's blue uptake and SEM and there was also an increase in blood cholesterol, glucose, and low-density lipoprotein levels.¹⁸ These experiments suggest that nicotine itself could be implicated in the development of atherosclerosis through endothelial damage as well as its known effect on the blood pressure and the plasma cholesterol level.

The concentration of CO in cigarette smoke is approximately 400 ppm, eight times the maximal permitted atmospheric level in industry.¹⁹ Atmospheric pollution, primarily from gasoline combustion, causes COHb levels of between 1.4% and 3.0% in nonsmoking London taxi drivers.²⁰ There is a close

correlation between the number of cigarettes smoked and COHb levels. Levels of 12% are common in heavy smokers and levels of 20% have been recorded in pipe smokers who inhale.²¹

In humans, COHb levels show a strong association with the development of the clinical features of atherosclerosis. In a study of 950 Danish smokers,²² subjects with a COHb level of 5% or more had a twenty-one times greater incidence of ischemic heart disease than those with a COHb level less than 5%. Despite these clinical correlations, animal experiments have failed to demonstrate an atherogenic effect of CO when given alone. Hugod et al.²³ exposed 60 rabbits to CO at a concentration of 200 ppm for 0.5 to 12 weeks and could find no significant histologic differences between the intima of the control and CO-exposed group, perhaps because 60% of the arteries in both groups showed intimal changes. Therefore they exposed a smaller number of animals to much higher concentrations of CO (2000 to 4000 ppm) but again were unable to show any histologic differences between the test and control arteries. They concluded that CO does not produce any morphologic changes in the arterial intima. Conversely other animal studies^{24,25} have shown an increase in the cholesterol content of the arterial wall after exposure to CO in the rabbit and also in primates.²⁶ But similar results could also be obtained by inducing hypoxia (10% to 16% inspired oxygen) for 8 weeks.²⁷

Therefore it is possible that CO exerts at least some of its effect as a result of tissue hypoxia, which is produced first by a reduction in the O₂Hb levels and second by shifting the oxygen dissociation curve to the left. The clinical evidence linking COHb levels and, by inference CO, with vascular disease is convincing but the animal experiments are inconclusive.

In a recent reappraisal of the "response-to-injury" hypothesis, Ross² has suggested that at least two pathways may lead to the intimal smooth muscle proliferation, which he still considers to be the key event in the development of atheroma. The first pathway suggests that monocytes and platelets interact with an intact endothelium to cause the release of growth factors, which promote smooth muscle proliferation at the same time as lipid-laden monocytes migrate through the interendothelial junctions to form fatty streaks.²⁸ The second pathway suggests that the direct stimulation of the endothelial cells causes them to release a growth factor that causes smooth muscle proliferation or a chemotactic factor that stimulates the migration of smooth muscle cells into the intima. Both hypotheses assume that the endothelium can be damaged without desquamation or cell loss and that this damage alters the permeability of the endothelial barrier.

The purpose of our experiments was to investigate the effects of cigarette smoke and two of its major components, nicotine and CO, on the permeability of the arterial wall to test the hypothesis that cigarette smoke increases the permeability of the arterial wall and that this might, over the long term, cause increased deposition of fibrinogen and lipids in the subintimal layers.

The four groups of dogs were all similar in terms of weight, arterial diameter, and anesthetic requirements. There was a small difference in the total duration of the experiments, the experiments of the smoke inhalation group taking longer. This extra time was accounted for by the need to increase blood COHb levels before the permeability measurements were started but should not have affected the measurement of fibrinogen transport across the arterial wall.

The mean arterial pressures in each group were different. The control, smoke inhalation, and CO groups maintained stable arterial pressure with less than 5% variation. The animals given nicotine had an increase in mean arterial pressure of 14% (range 9.8% to 21%) during the first 2 hours of the nicotine infusion. The pressure then decreased slightly but remained higher than the resting value. There was also a 25% increase in the pulse rate concurrent with the increase in pressure. These changes reflect the

physiologic effects of nicotine.²⁹ The control and nicotine groups also had slightly higher mean arterial pressures during the permeability experiments than the smoke inhalation and CO groups. However, both these changes would have marginally increased the transport of the fibrinogen across the arterial wall⁴ but both the control and nicotine groups had significantly lower PCs than the smoking and CO groups.

The smoke inhalation and the CO groups had a significantly greater permeability than the control group and there was no significant difference between the control group and the nicotine group. Therefore it would appear that cigarette smoke does increase the flux of fibrinogen across the arterial wall and this may be one of the mechanisms by which cigarettes exert their atherogenic effect. The effect appears to be caused by the CO. If cigarette smoke and CO are increasing the permeability of the arterial wall to ¹²⁵I-labeled fibrinogen, how is this change brought about? EM and LM revealed no evidence of endothelial damage nor were there any signs of an inflammatory infiltrate. This suggests that neither the capsule nor the cigarette smoke had significantly injured the length of artery being studied. We would have expected an inflammatory response to develop within 90 minutes of any significant injury.³⁰ However, although no morphologic changes have been demonstrated in the artery as a result of having a capsule placed around it, it seems likely that the capsule must disturb the artery at a subcellular level and this must be accepted as a limitation of the model.

Transmission electron microscopy of the intracapsular artery also failed to show evidence of arterial wall damage and in particular there was no sign of subendothelial edema. The short duration of exposure of the animals to cigarette smoke and two of its components may mean that the morphologic changes described by Woolf and Wilson Holt³ have not had time to develop. However, the permeability of the arterial wall began to increase almost immediately after the exposure to cigarette smoke and CO. The mechanism producing this change is not clear but it seems likely that there must be subcellular changes altering transmural transport.

In both the smoke inhalation and CO groups the COHb levels were elevated throughout the experiment and there was a correlation between the PC and the blood COHb level in both groups. There was also a concomitant decrease in the %O₂Hb. The mean decrease in the smoke inhalation group was only 7%, but the mean decrease in the CO group was 16%. Therefore hypoxia may be playing a part in altering arterial wall permeability. Kjeldsen et al.³¹

have demonstrated an increase in the cholesterol content of rabbit aorta that has been kept hypoxic (10% inspired O₂) for 8 weeks. Unfortunately we only measured the %O₂Hb at the beginning of the control group study because we did not anticipate any significant changes; however, it would have been interesting to know the %O₂Hb levels throughout the control study. It is quite likely that there was a slight decrease in %O₂Hb and we intend to investigate whether mild hypoxia alone increases arterial permeability in the model.

The risk factors associated with an increased incidence of atherosclerosis include hypertension, hyperlipidemia, diabetes, and smoking. The mechanism by which these risk factors might initiate or accelerate the development of atheroma is not known, but from the results of these experiments it can be postulated that smoking upsets the balance between the insudation of plasma macromolecules into the arterial wall and their subsequent clearance by altering the permeability of the endothelium. The clearance mechanisms of the arterial wall may not then be able to adequately remove these molecules, thereby resulting in an intramural accumulation of lipids and fibrinogen.

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DISCUSSION

Dr. David J. Effeney (Brisbane, Australia). Last year at this meeting I presented some data on work that I have done on nicotine and carbon monoxide. The reason that people smoke is nicotine, named after a Frenchman, Jean Nicot. It is clear that nicotine is the substance in cigarette smoke to which people are addicted.

When you study humans who smoke cigarettes, it is important to realize that once you give nicotine or smoke a cigarette you get a rapid increase in plasma nicotine concentration to approximately 20 to 25 ng/ml. Most people set their levels in that type of range. Depending on whether you have alkaline or acid urine, you eliminate nicotine at a faster or slower rate. People then smoke to maintain the nicotine concentration.

There are some studies in humans who smoked high- and low-nicotine cigarettes. In controlled situations the nicotine concentration rapidly plateaus at about 40 ng/ml with high-nicotine cigarettes, whereas with the low-nicotine cigarettes, this occurs at about 10 ng/ml. When low-nicotine cigarettes are smoked, you get many more of the products of pyrolysis and higher carboxyhemoglobin levels than when high-nicotine cigarettes are smoked.

We have shown that prostacyclin production was altered by the infusion of nicotine and exposure to carbon monoxide. The nicotine decreased it, carbon monoxide increased it. We have a differential effect of the two components in the gaseous and the solid phase of cigarette smoke. However, when we looked at nicotine plus carbon monoxide—and I will ask Professor Browse to tell me whether he is planning to study the combination of nicotine and carbon monoxide—there was a further increase in the amount of prostacyclin produced by the endocardial cells.

Carbon monoxide is interesting. It causes cellular hypoxia, and there is a change to anaerobic metabolism, at 12% carboxyhemoglobin. There are metabolic changes in other tissues, about 20%, and of course carbon monoxide also binds to other heme-containing enzymes. There is a direct toxic effect that perhaps alters the permeability of the cell.

I have four other questions. The experiments were done at the time fibrinogen would be redistributing after injection. Did you think that affected your results? Why was the anesthetic requirement increased about three times in the carbon monoxide-treated dogs? You used a CO oximeter to measure the blood; at least in rabbits you cannot use a CO oximeter to measure carboxyhemoglobin because it gives you an incorrect value. I wondered whether you had to change the filters to get an accurate measure? Finally, how do you propose to proceed to determine the contribution of the various parts of the arterial wall to macromolecular movement?

Professor Browse. Dr. Effeney, we do intend to combine nicotine and carbon monoxide to see whether the combination has any enhancing effect. Hypoxia did occur

but we did not measure it throughout every experiment. The oxyhemoglobin levels decreased by approximately 7% in the smoke inhalation group but by 12% in the CO group. Although relatively small, this change may have altered permeability. We intend to perform a study on the effect of hypoxia.

The anesthetic requirement was not increased in the carbon monoxide group; there was no statistical difference in any of the groups regarding anesthetic requirement. The CO oximeter had been adapted to measure blood COHb in dogs.

Dr. James O. Menzoian (Boston, Mass.). Professor Browse, we have been very interested in changes in arterial wall and its permeability. In fact, last year at this meeting we presented our own data in which we showed that as a result of cholesterol feeding there was a marked enhancement of permeability across the rabbit carotid artery. Any study that can show changes in permeability to macromolecules I think is critical to our understanding of atherosclerosis.

Your model involves at least a local dissection of the artery. We found in our own model that any manipulation of the artery was extremely detrimental to the integrity of the wall. Please comment on disruption of local lymphatics and the vasa blood supply and the effect you think that might be having on permeability changes.

Second, you showed that the arterial wall counts also increased. Did you find that there was a direct correlation, that is, as the wall counts went up, did the counts in your fluid chamber parallel this increase?

Third, you express your data as standard error of the mean, and we know that in fact the data are tight, suggesting an even distribution and parametric data. I wondered whether analysis of these data would be best done by a *t* test or analysis of variance. Why did you use the Mann-Whitney test? I thought that would be better for nonparametric data.

Finally, you have shown an interesting finding that there was enhanced permeability to a large molecule as a result of cigarette smoking, yet you have shown by the histology studies that the wall was intact and the integrity of the endothelium looked good. Would you speculate on the origin of this enhanced permeability?

Professor Browse. Dr. Menzoian, we have performed a series of studies, which I am afraid have not been published, on the effect of dissecting the artery, and on removing the vasa vasorum. Removing the vasa vasorum does have a significant effect; it actually increases transport through the wall. At one stage we were concerned that the isotopes present in the capsule were getting there via the vasa vasorum and not through the whole arterial wall. The increased permeability without the vasa vasorum refuted this notion. You cannot put the capsule around the artery without damaging it somewhat; one just tries to do it with minimal trauma. We have performed some uptake studies in arteries that have not had capsules around them and

they correlated well with the studies on arteries within the capsule. There has been a positive correlation between wall counts and the amount of isotope that gets into the capsule.

We did perform analyses of variance as well as Mann-Whitney and *t* tests and they all showed the same degrees of significance.

We do not know what the mechanism is. If you continue to give cigarette smoke or carbon monoxide you do see electron microscopic changes in endothelial cells. Perhaps we are seeing the beginning of this process before the cell damage is visible.

We have already started some studies with the use of dogs' labeled lipids and find the same sort of effect on lipid transport with some changes in the nature of the lipid as they pass through the wall.

Dr. John Hines Kennedy (London, England). I wish to comment on our data from studies still in progress at the Hôpital Lariboisière in Paris where Dr. Alain Tedgui and I have been studying what is usually thrown away: the button of ascending aorta that comes out when vein bypass graft for coronary disease is done. Most of the patients had already given up smoking.

If these buttons were incubated in a mixture of two labels, ¹²⁵I-labeled serum albumin and ⁵¹Cr-labeled EDTA for 2 hours, there was a twofold difference in range of the ratio of distribution volumes of the two isotopes when sampled from the same patient at the same time; I think that makes interpretation difficult. This is a complicated problem.

Professor Browse. The original objective when designing this capsule was to use it in humans. Unfortunately, it needs to be on for 2 or 3 hours and one does not usually leave an artery exposed for that long during an arterial operation, while doing nothing to it so we have not yet had a chance to use it in humans. We hope to be able to put it around some femoral arteries while doing the top end of an aortofemoral bypass. We have investigated different arteries in the dog and not found any differences but of course, these are all healthy arteries.

Dr. Wesley S. Moore (Los Angeles, Calif.). I would like to add some evidence that I believe may be complementary to the data that you presented.

In 1978, at an international meeting in Prague, there were two papers presented back to back that I found very exciting. The first involved the sampling of peripheral venous blood in smokers and comparing it to that in non-smokers, looking for the presence of circulating endothelial cells. Indeed, circulating endothelial cells could be found in high levels in smokers. This suggested that a type of endothelial desquamation was taking place in patients who were exposed to cigarette smoke.

The second presentation was an experimental study of arteries from animals that had been exposed to tobacco smoke inhalation. The investigators studied the endothelial surfaces and found that one could see specific areas of crypt formation representing endothelial denudation. To determine whether or not this was more than happenstance, we took this back to the laboratory a few years later. Dr.

George Seipert, one of my fellows, and I exposed groups of rats to tobacco smoke in smoke chambers. Although a rat is smart enough not to smoke, if you place them in a chamber in which tobacco smoke is ventilated into the chamber, you can produce the same systemic effect by altering the ambient environment: an observation that was later taken up by restaurants in Beverly Hills.

In exposing these animals to smoke 15 minutes per day three times a day and then examining their arterial surfaces, one could indeed see definite areas of crypt formation where the endothelial cells were gone in contrast to the control animals. Perhaps these are the holes, if you will, through which the fibrinogen is passing.

I am sympathetic with your concept that we should not make a forceful statement that we know that this causes arteriosclerosis. On the other hand, I think if we equivocate in our comments to our patients they tend not to respond to casual suggestion or advice; therefore I tell them that smoking is absolutely devastating to their arterial tree and they should stop it. Most patients think that smoking only causes lung cancer and they fail to recognize the important relationship between tobacco smoke and arterial disease. We usually get fairly good compliance in this way, and, in some of our female patients who are still noncompliant, the final coup de grâce occurs if you tell them that smoking causes wrinkles; this works every time.

Dr. George Johnson, Jr. (Chapel Hill, N.C.). We North Carolinians appreciate Dr. Browse and the British subtle support of our endeavors. They were on our side in the Civil War, although this was not overt. It is interesting that this presentation, although not overt, may show that tobacco smoking is not so bad (relatively) after all. I am not sure Professor Browse meant it that way. However, this study did not have a control for smoking. The study results showed that nicotine was not bad but that carbon monoxide was. Perhaps if we were to "smoke" some Los Angeles air or "corn silk" or "rabbit tobacco" this would have the same effect as the carbon monoxide. Is tobacco any worse?

I guess we North Carolinians, and others—not myself because I think smoking is bad for you, and I don't smoke—can go back to Robert Ripley, who published the creed of Richard Dean, Stanley Crawford, Bill Evans, and others when he said, "Tobacco is a filthy weed, I like it. It satisfies no normal need, I like it. It makes you thin, it makes you lean, it takes the hair off your bean. It's the worst damn stuff I've ever seen, but I like it."

Professor Browse. Well, of course, we did have control studies and there was a significant difference between them and the smokers and in those given carbon monoxide. Interestingly, the carbon monoxide levels of the London taxi driver who does not smoke ranges between 1.5% and 3% and the carbon monoxide level of the London greyhound is about 1.5%. So living in London is probably rather worse than living in North Carolina, even if you're a nonsmoking dog.

Mr. H. H. G. Eastcott (London, England). I have an anecdotal report of a patient with a 25-year follow-up. He

is a Roman Catholic priest. I amputated both of his legs in 1960. I said, "Now, Father Anthony, swear that you will stop smoking." He said, "I swear, Dr. Eastcott, that I'll stop smoking." I saw him again about 2 years ago. Father Anthony—shorter, of course; "Cut off my legs and call me 'Shorty Anthony,'" they call him now—but I said, "How about it, did you remember your promise to the Almighty?" He said, "Yes, I haven't smoked since, but the snuff is wonderful!"

Now, this supports completely the results of Professor Browse. It's the conflagration of tobacco that does the damage and not just the ingestion of nicotine.

Professor Browse. I cannot cap Mr. Eastcott's story but I would like to remind you that all this work was done by Mr. David Allen and not by myself. Thank you all for your kind and stimulating comments.

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